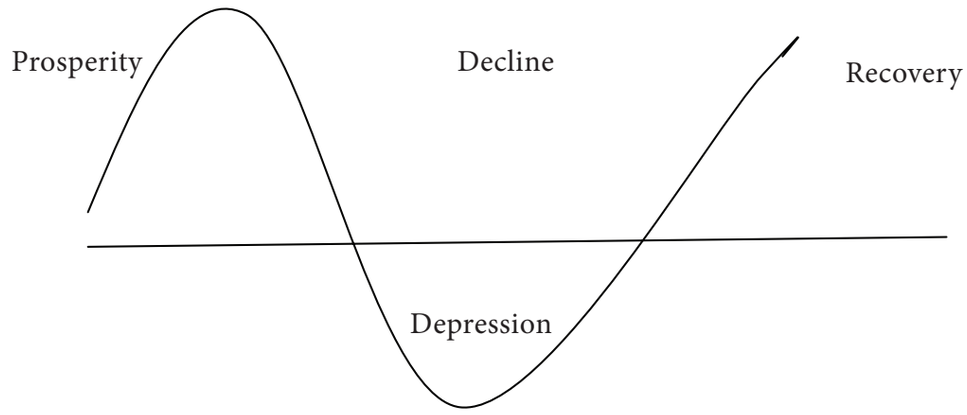
(iii) Cyclical variation

Cyclical variations refer to periodic movements in the time series about the trend line, described by upswings and downswings. They occur in a cyclical fashion over an extended period of time (more than a year). For example, the business cycle may be described as follows.



The cyclical pattern of any time series tells about the prosperity and recession, ups and downs, booms and depression of a business. In most of the businesses there are upward trend for some time followed by a downfall, touching its lowest level. Again a rise starts which touches its peak. This process of prosperity and recession continues and may be considered as a natural phenomenon.

YOU WILL KNOW

Cyclic movements are mainly due to Trade cycle.

(iv) Irregular variation

In practice, the changes in a time series that cannot be attributed to the influence of cyclic fluctuations or seasonal variations or those of the secular trend are classified as irregular variations.

NOTE

There is no statistical technique for measuring or isolating irregular fluctuations

In the words of Patterson, "Irregular variation in a time series is composed of non-recurring sporadic (rare) form which is not attributed to trend, cyclical or seasonal factors".

Nothing can be predicted about the occurrence of irregular influences and the magnitude of such effects. Hence, no standard method has been evolved to estimate the same. It is taken as the residual left in the time series, after accounting for the trend, seasonal and cyclic variations.

YOU WILL KNOW

Irregular variation is also called erratic fluctuations.

7.4 FORECASTING

The importance of statistics lies in the extent to which it serves as the basis for making reliable forecasts, against arbitrary forecast with no statistical background.

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